

Semi-Supervised Learning with Max-Margin Graph Cuts

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Archival repo-owned poster render reconstructed from the historical poster and AISTATS 2010 paper

Motivation

- Semi-supervised max-margin learning should use abundant unlabeled data without turning the optimization problem non-convex.
- Manifold regularization of SVMs is a strong baseline, but with severely restricted discriminators—especially linear functions—it can produce unintuitive decision boundaries.
- We want a method that first exploits graph structure and then learns a max-margin classifier consistent with that structure.

Key idea

Use the harmonic-function solution to infer soft labels on the graph, then train a max-margin discriminator conditioned on those induced labels.

Two-stage construction

1. Graph inference

Build a data-adjacency graph and compute a regularized harmonic-function solution. This propagates information from the labeled examples through nearby unlabeled points.

2. Max-margin projection

Learn a discriminator that maximizes margin with respect to the labels induced by the graph solution. Increasing the graph regularization smoothly interpolates between semi-supervised learning on all data and ordinary supervised learning.

- Convex optimization problem
- Kernel trick available
- Sparse graph adjacency can be exploited computationally

Why it differs from prior approaches

Versus semi-supervised SVMs with a hat loss

The proposed objective is convex and can be solved optimally, rather than relying on a non-convex unlabeled-data loss.

Versus manifold regularization

The graph solution is used to define the classification task before imposing the max-margin discriminator. This is particularly useful when the hypothesis class is strongly constrained.

Theory

The generalization error can be controlled by combining transductive graph-learning bounds with standard inductive margin bounds.

Favorable properties

- Convex objective: optimization has a global solution.
- Kernelizable: nonlinear discriminators require no explicit feature expansion.
- Graph-aware: unlabeled examples influence the induced labels through connectivity.
- Sparse-friendly: sparse adjacency matrices can be used directly.
- Generalization bounds: the construction admits a theoretical analysis.

Experimental setup

- Three UCI tasks: letter recognition, digit recognition, and image segmentation.
- Linear, cubic, and RBF kernels.
- Labeled fractions from 1% to 10%.
- Compare supervised SVM, manifold regularization (MR), and max-margin graph cuts (GC).

Qualitative result

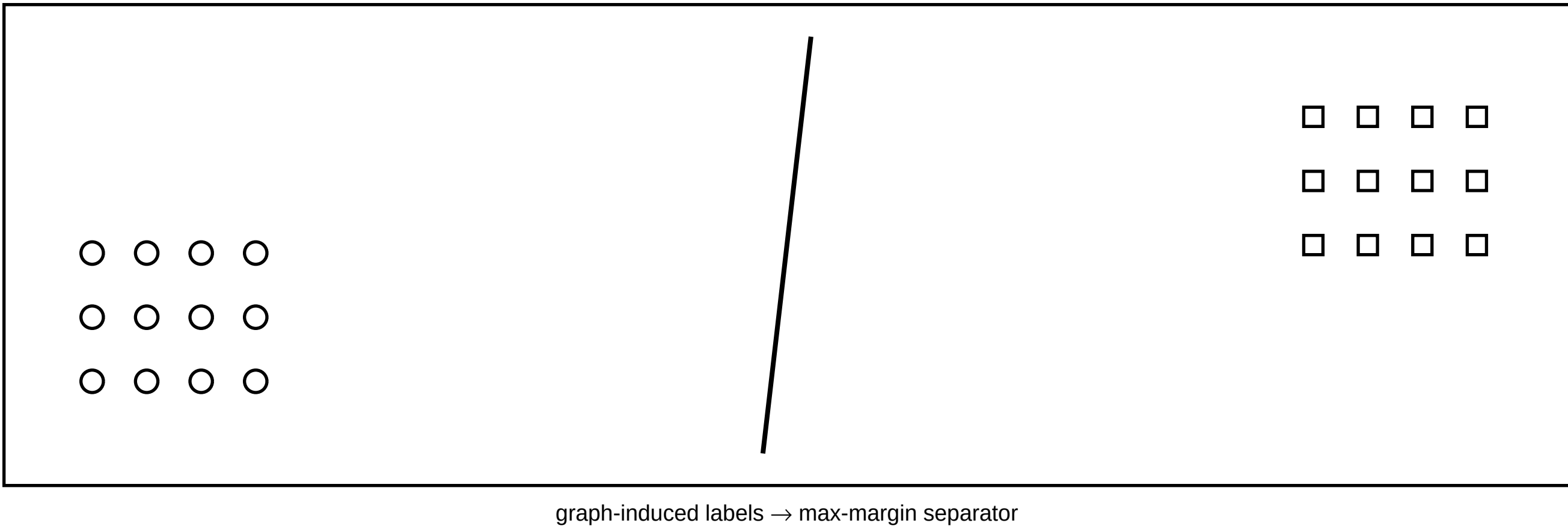
GC is especially attractive for restricted linear decision boundaries; with flexible RBF kernels, GC and manifold regularization become much closer.

Algorithm sketch

- ➊ **Input**
labeled examples + unlabeled examples + graph W
- ➋ **Harmonic step**
solve the regularized graph-label propagation problem
- ➌ **Induced targets**
obtain soft/graph-consistent labels for all examples
- ➍ **Margin step**
fit a maximum-margin discriminator to the induced labels
- ➎ **Predict**
use the learned discriminator on new examples

Geometric intuition

Graph propagation separates two dense groups even when a linear manifold-regularized SVM cannot express the desired boundary directly.



Empirical summary

Across the historical poster’s 36 UCI comparisons, max-margin graph cuts outperform manifold regularization in 29 experiments.

The strongest gains appear when the classifier class is too restricted for manifold regularization to faithfully reflect the cluster structure.

Letter recognition	GC usually best
Digit recognition	GC competitive/best
Image segmentation	GC often best
RBF kernels	GC = MR

Takeaways

- Separate “use the graph” from “fit the classifier.”
- This separation keeps the optimization convex and makes the method robust when the final discriminator is simple.
- The work is an early example of combining graph-based semi-supervision with margin-based prediction in a modular way.